

31 March - 4 April 2025

Time: 20 min

8

Surname:

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Initials:

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Student number:

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M-number:

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Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator.

QUESTION 1 [1, 2, 1]

Consider the function

$$f(x) = \left(\frac{1}{x} - \frac{1}{|x|} \right) \sin x.$$

Determine

(a) $\lim_{x \rightarrow 0^+} f(x).$

$$|x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right) \sin x$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x} \right) \sin x = 0$$



$$|x| = -x$$

(b) $\lim_{x \rightarrow 0^-} f(x).$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right) \sin x = \lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x} \right) \sin x, \text{ since } |x| = -x$$

$$= \lim_{x \rightarrow 0^-} \frac{2 \sin x}{x} = 2 \lim_{x \rightarrow 0^-} \frac{\sin x}{x} = 2 \times 1 = 2.$$

(c) $\lim_{x \rightarrow 0} f(x).$

Does not exist.

QUESTION 2 [1.5, 1.5]

Consider the function

$$g(x) = \frac{x^2 + 2x + 1}{1 - x^2} = \frac{(x+1)(x+1)}{(1-x)(1+x)} = \frac{x+1}{1-x}$$

(a) Find the vertical asymptote(s) of g , if any.

$$\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} \frac{x+1}{1-x} = \infty$$

$$\begin{aligned} x &\rightarrow 1^- \\ \Rightarrow x &< 1 \\ \Rightarrow x - 1 &< 0 \\ \Rightarrow 1 - x &> 0 \end{aligned}$$

NB
 $\Rightarrow x = 1$ is a vertical asymptote of g .

(b) Find the horizontal asymptote(s) of g , if any.

$$\begin{aligned} \lim_{x \rightarrow \infty} g(x) &= \lim_{x \rightarrow \infty} \frac{x+1}{1-x} \left(\frac{\infty}{-\infty} \right) \\ &= \lim_{x \rightarrow \infty} \frac{1 + 1/x}{1/x - 1} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} g(x) &= \lim_{x \rightarrow -\infty} \frac{x+1}{1-x} \left(\frac{-\infty}{\infty} \right) \\ &= \lim_{x \rightarrow -\infty} \frac{1 + 1/x}{1/x - 1} \\ &= -1 \end{aligned}$$

$\Rightarrow y = -1$ is a horizontal asymptote.

QUESTION 3 [1 mark]Use the definition of the "derivative at a point" to find the derivative of the function $f(x) = 2x - 3$ at $x = 2$.

$$f'(2) = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$$

$$f(2) = 4 - 3 = 1$$

$$\begin{aligned} &= \lim_{x \rightarrow 2} \frac{2x - 3 - 1}{x - 2} = \lim_{x \rightarrow 2} \frac{2(x - 2)}{(x - 2)} \\ &= 2. \end{aligned}$$